Hardware Implementation

The Smart Farm Integration system represents a sophisticated fusion of Internet of Things (IoT) hardware and centralized software analytics, designed to transform raw environmental data into actionable agricultural intelligence. At the heart of this ecosystem lies the ESP32 microcontroller, which serves as the primary gateway, translating complex electrical signals from specialized sensors—ranging from capacitive soil moisture probes to non-invasive current transformers—into standardized JSON payloads. By leveraging high-speed Wi-Fi connectivity and robust API protocols, the system creates a seamless feedback loop where real-time metrics are visualized on a web-based dashboard, allowing for precision monitoring of nutrient levels, water availability, and energy consumption. This integration not only bridges the gap between the field and the screen but also empowers the dashboard to act as a command center, remotely triggering automated irrigation and lighting systems to optimize crop yields and operational efficiency.

Temperature and Humidity

To detect environmental conditions, the ESP32 is integrated with a **DHT22** digital sensor, which measures both temperature and relative humidity. The physical connection is streamlined: the VCC connects to the 3.3V pin on the ESP32, the GND connects to the ESP32 GND, and the Data Pin is linked to GPIO 4. This single data line carries both readings to the ESP32, which acts as a translator, converting raw electrical signals into a standardized JSON format. Because the dashboard is web-based, the ESP32 utilizes its onboard Wi-Fi to "POST" these combined metrics—temperature and humidity—to a specific API endpoint for real-time visualization.

To ensure the accuracy of the data reflected on your dashboard, physical placement of the hardware is critical. You should position the sensor in a well-ventilated area at approximately chest height to capture representative ambient conditions, avoiding "dead zones" like corners or sealed enclosures where heat and moisture can become trapped. It is vital to keep the sensor away from direct heat sources—such as sunlight, radiators, or the ESP32's own voltage regulator—to prevent "false high" temperature readings and artificially low humidity values.

Additionally, ensure the device is mounted on a stable, vibration-free surface with jumper wires kept under 30cm to prevent data packet loss or signal jitter. Finally, verify the placement allows for a strong Wi-Fi signal to prevent "Sensor Offline" errors on your dashboard, and consider a brief side-by-side calibration with a standard thermometer and hygrometer to account for any consistent offsets in your digital visualizations.

Figure 1: DHT22 Digital Sensor

Total electricity

To detect the total electricity usage of the entire farm, we paired ESP32 with **Non-Invasive Current Transformer (CT) Sensor, which is SCT-013.** The SCT-013 sensor acts like a clamp that snaps around the main "Live" wire of the farm electrical panel. The sensor is connected to an ADC (Analog-to-Digital Converter) before reaching the ESP32, then ESP32 needs to convert the analog magnetic readings into a digital "Amperage" value. Once the ESP32 has the Amperage, it uses a simple mathematical formula to calculate total power:

$$Power(Walts) \ = \ Voltage\ (240V) \backslash\ times\ Current\ (Amps)$$

To ensure accuracy and safety, the sensor must be placed exclusively around the single main live wire within the distribution box, housed in an insulated enclosure to prevent interference. This allows the dashboard to provide high-precision visualizations of energy trends, helping to identify peak usage periods across the entire farm operation.

Figure 2: Non-Invasive Current Transformer (CT) Sensor: SCT-013

Total volume of water in tank

To monitor the water level in the farm's storage tank, the ESP32 uses an **HC-SR04 ultrasonic sensor** mounted at the top of the tank. The VCC connects to the 5V pin (for better range) or 3.3V, and GND connects to ESP32 GND. The Trig pin (Trigger) and Echo pin are connected to two GPIOs (e.g., GPIO 5 and GPIO 18). Then, the ESP32 acts as the translator by measuring the duration of the echo pulse and applying a mathematical formula :

$$Distance \ = \ (Time\ x\ 0.034)\ /\ 2)$$

Then, ESP32 formats it into JSON, and transmits it via Wi-Fi to our dashboard. For accurate visualization, the sensor mounted centrally at the top of the tank to avoid wall interference and kept away from the direct splash of inlet pipes. This setup allows the dashboard to provide real-time updates on water availability and trigger automated alerts or pump controls when levels reach critical thresholds.

Figure 3: HC-SR04 ultrasonic sensor

Soil moisture

To detect soil moisture, the ESP32 is integrated with a **Capacitive Soil Moisture Sensor (v1.2)**, which is preferred over older resistive models because it resists corrosion when buried in damp soil. The physical connection involves connecting the VCC to the 3.3V pin, GND to ESP32 GND, and the Analog Output (AUOUT) to an Analog-to-Digital Converter pin, such as GPIO 34. The ESP32 acts as a translator by reading the varying voltage levels—where higher voltage typically represents dry soil and lower voltage represents wet soil—and mapping these values into a 0–100% moisture scale. This data is then formatted into JSON and pushed via Wi-Fi to your web-based dashboard.

For our dashboard to provide actionable insights, the physical placement of the sensor is vital. The probe should be inserted vertically into the soil near the root zone of the plants, ensuring the "electronic" top part of the sensor remains above the soil and clear of irrigation spray to prevent short circuits. Also, we need to ensure that we press the soil firmly around the probe to eliminate air gaps, which might cause erratic "jitter" in your dashboard graphs. Additionally, since soil mineral content varies, we perform a "dry" and "wet" calibration: record the sensor value in open air (0%) and in a cup of water (100%) to ensure the percentage shown on your software is accurate for your specific farm soil. This setup allows the dashboard to display real-time hydration levels and can be used to trigger automated irrigation systems only when the soil truly requires water.

Figure 4: Capacitive Soil Moisture Sensor (v1.2)

Electrical Conductivity

Electrical Conductivity is to measure the nutrient concentration in the water. To monitor the nutrient levels in your farm's irrigation or hydroponic system, the ESP32 is integrated with an **Electrical Conductivity (EC) Sensor**. The physical setup involves an EC probe connected to a signal conditioning board, which is then wired to the ESP32: VCC to 5V, GND to ESP32 GND, and the Analog Output to an analog pin like GPIO 32 or 35. The ESP32 acts as a translator by measuring the voltage output—which increases as the water's mineral and salt content (nutrients) becomes more conductive—and converting these raw signals into a measurement of millisiemens per centimeter (mS/cm). This data is then packaged into a JSON format and transmitted via Wi-Fi to your web-based dashboard.

For the dashboard to accurately reflect the nutrient health of your crops, physical placement and maintenance are critical. The probe should be submerged in an area of the tank or pipe with consistent water flow to ensure it is measuring a well-mixed solution rather than a stagnant pocket of concentrated fertilizer. It is essential to keep the probe away from the sides of the tank and metal components that might interfere with the electrical field. Because EC readings are highly sensitive to temperature, most systems use "Temperature Compensation" (often pairing this with the DHT22 or a DS18B20 sensor) to ensure the dashboard data remains stable regardless of weather changes. Regular cleaning of the probe tips is required to prevent "bio-fouling" or mineral buildup, which can cause the nutrient levels to appear lower on your software than they actually are. This integration allows the dashboard to provide a real-time "nutrient profile," helping you maintain the perfect balance for plant growth.

Figure 5: Electrical Conductivity (EC) Sensor

Light_intensity

To detect whether the farm lights are operational and to measure their brightness, the ESP32 is integrated with a **BH1750 Digital Light Sensor**. BH1750 communicates via the I2C protocol, connecting the VCC to 3.3V, GND to GND, SCL to GPIO 22, and SDA to GPIO 21. The ESP32 acts as a translator by reading the light intensity in Lux, which it then maps to a 0–100 level scale. To determine the "ON/OFF" status, the firmware applies logic: if the intensity is above a specific threshold (e.g., 10 Lux), the status is set to "ON." This data is packaged into JSON—including the id, status, and level—and sent via Wi-Fi to your web-based dashboard.

Proper physical placement is vital to ensure the dashboard reflects the true state of your lighting system rather than environmental interference. The sensor should be positioned with a direct "line of sight" to the light source it is monitoring, but it should be shielded from external sources like sunlight or passing vehicle headlights to avoid "false ON" readings during the night. If you are monitoring indoor grow lights, mounting the sensor at the plant canopy level is best to measure the actual light reaching the crops. Additionally, ensure the sensor lens is kept clean; dust or debris accumulation will cause the light level to appear lower on your dashboard over time. This integration allows your software to confirm that lights are functioning as scheduled and provides the data necessary to adjust brightness levels for energy efficiency or plant health.

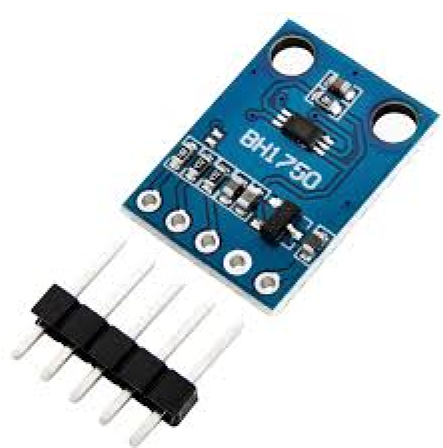

Figure 6: BH1750 Digital Light Sensor

Sprayer System

To control and monitor the **sprayer** in your farm system, the ESP32 is integrated with a **5V Relay Module** and a **Solenoid Valve**. The physical connection involves connecting the ESP32's 3.3V and GND to the relay's power pins, while the Relay Input (IN) pin is connected to a digital pin such as GPIO 13. The relay acts as an electrically operated switch, allowing the low-power ESP32 to safely control the high-power circuit of the sprayer. The ESP32 acts as the controller; when it receives a command from your dashboard, it pulls the GPIO pin "HIGH" or "LOW" to open or close the valve. This operational state is then formatted into a JSON payload and sent via Wi-Fi to update your web-based dashboard in real-time.

For the dashboard to provide effective control, the physical placement of the sprayer hardware must be carefully managed. The relay and ESP32 should be housed in a waterproof NEMA-rated enclosure located away from the actual spray nozzles to prevent short circuits caused by mist or overspray. The sprayer nozzles themselves should be positioned to ensure even coverage across the rack or plants, with the tubing securely fastened to prevent "pressure kicks" from loosening the connections. It is important to include a "manual override" button or a failsafe in your software logic—such as a timer that automatically turns the sprayer off after 60 seconds—to prevent flooding if the Wi-Fi connection drops while the sprayer is active. This integration allows your dashboard to serve as a remote command center, enabling you to trigger hydration or pest control manually or via automated schedules based on the sensor data collected from your other units.

Figure 7: 5V Relay Module

Figure 8: Solenoid Valve